

Supplementary Information

The Atlantic is freshening for opposing reasons

This document collects supplementary material that complements—but does not duplicate—the Methods section of the main text. We give: (i) a model-by-model audit of the ocean numerical infrastructure and a demonstration that no single ocean parameterization covaries with the mechanism class (Section and Table 2); (ii) the per-figure CMIP6 ensemble bookkeeping, since different downstream analyses retain different subsets of the 15-model decomposition ensemble (Section); (iii) the rationale for excluding the SODA 3.15.2 1998 annual mean as a data-corruption outlier (Section); and (iv) supporting figures S5–S9.

Model selection criteria for the CMIP6 ensemble

The CMIP6 ensemble used in the main text was assembled by applying a single set of pre-registered inclusion criteria to the full federated ESGF archive (DKRZ, LLNL, IPSL, BADC, CEDA, NCI mirrors, queried via the DKRZ Solr index with `distrib=true` on 2026-04-28). A model is included if and only if it publishes:

1. monthly meridional ocean velocity (`vo`) and salinity (`so`) (CMIP6 table `0mon`);
2. both the `historical` and the `ssp585` experiment;
3. realisation `r1i1p1f1` (the most widely available ensemble member; we use the same realisation in every model to remove member-selection ambiguity);
4. native ocean grid (`grid_label=gn`). RegridDED archives (`gr`) are excluded because regridding distorts the mass-velocity-salinity coupling that the F_{ovs} decomposition resolves.

A model is excluded a priori, regardless of `vo/so` publication status, if its ocean component uses isopycnal (density) vertical coordinates (NorESM2-LM, NorESM2-MM): the depth-coordinate remap that would be needed to evaluate $\int v[\bar{S} - S_0] dz$ on these grids would itself reshape the velocity-vs-salinity partition we are trying to diagnose.

Models in the analysed ensemble. Twenty-five CMIP6 models survive the criteria for the `historical` + `ssp585` decomposition. The original sixteen-model core (ACCESS-CM2, CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, FGOALS-g3, FIO-ESM-2-0, GFDL-CM4, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL) was extended in the v4.3 federated-ESGF + Pangeo discovery sweep with nine additional native-grid models that publish both the required experiments at `r1i1p1f1`: ACCESS-ESM1-5, BCC-CSM2-MR, CESM2-WACCM, CIESM, CMCC-ESM2, EC-Earth3, FGOALS-f3-L, MRI-ESM2-0, TaiESM1. Of the 25, $n = 20$ exhibit forced AMOC weakening (see Section); the headline mechanism partition is calculated on this $n = 20$ subset, while the AMOC26N mechanism-conditional weakening (Main Fig. 2b,c) is restricted to the 16 models with available `msftmz` fields, leaving $n = 13$ forced-weakening models in the gap-quantification subset.

Models considered but excluded. Table 1 lists the 30 additional CMIP6 models queried against the federated ESGF index, with the precise reason each was dropped. The exclusions fall into four categories: (i) isopycnal ocean coordinates (NorESM2 family); (ii) native-grid `vo` or `so` not published for one or both required experiments; (iii) only regridDED archive available; and (iv) unstructured ocean mesh (AWI-CM-1-1-MR uses FESOM triangular finite-element grid; native `vo/so` are published, but section extraction at 34.5°S requires interpolation infrastructure beyond our NEMO/POP/MOM/MRI.COM-style depth-coordinate kernel). The exclusions are therefore archive-availability driven, not result-driven; the ensemble has not been pruned to favour a particular outcome.

Table 1: CMIP6 models considered for the headline ensemble but excluded, with the exact reason each was dropped (queried federated ESGF index 2026-04-28). Columns “vo h”, “so h”, “vo s”, “so s” report the number of native-grid (gn) files returned by the Solr index for monthly vo and so in the **historical** and **ssp585** experiments; a value of 0 in any column triggers exclusion. The reason column gives the dominant driver of the exclusion.

Model	vo h	so h	vo s	so s	Reason for exclusion
NorESM2-LM	–	–	–	–	isopycnal ocean coordinates
NorESM2-MM	–	–	–	–	isopycnal ocean coordinates
AWI-CM-1-1-MR	ok	ok	ok	ok	FESOM unstructured mesh
INM-CM4-8	0	0	0	0	no native vo/so on ESGF
INM-CM5-0	0	0	0	0	no native vo/so on ESGF
KACE-1-0-G	0	0	0	0	no native vo/so on ESGF
KIOST-ESM	0	0	0	0	no native vo/so on ESGF
IITM-ESM	0	0	0	0	no native vo/so on ESGF
MPI-ESM1-2-HAM	0	0	0	0	no native vo/so on ESGF
HadGEM3-GC31-MM	0	0	0	0	native files not on federated index
SAM0-UNICON	ok	ok	0	0	no ssp585 on ESGF
NorCPM1	–	–	0	0	no ssp585 on ESGF
GISS-E2-2-G	0	0	0	0	no native vo/so on ESGF
CNRM-ESM2-1	0	0	0	0	native files indexed only at IPSL
GFDL-ESM4	ok	ok	0	0	no native ssp585 vo/so
EC-Earth3-Veg	ok	ok	0	0	no ssp585 vo/so
GISS-E2-1-H	0	0	–	–	regridded only
GISS-E2-2-H	0	0	–	–	regridded only
E3SM-1-0	–	–	–	–	regridded only
E3SM-1-1	–	–	–	–	regridded only
E3SM-2-0	0	0	0	0	no native vo/so on ESGF
EC-Earth3-CC	–	–	–	–	near-duplicate of EC-Earth3 (same NEMO core)
EC-Earth3-Veg-LR	–	–	–	–	near-duplicate of EC-Earth3 (same NEMO core)

The discovery code that performed the federated Solr query (`scripts/discover_cmip6_expansion.py` in the public code repository) is reproducible against any future ESGF state; we encourage future revisions to re-run it as additional models or republished native-grid archives appear. The full per-model vo/so file-count grid against $(\text{historical}, \text{ssp585}) \times (\text{gn}, \text{gr})$ is preserved in `data/results/discovery_cmip6_expansion.csv`.

Ocean component reference per weakening model

Table 2 lists the ocean and atmospheric components of each CMIP6 weakening model, grouped by the mechanism class assigned from the centennial F_{ovs} decomposition. We provide this as an audit reference rather than a causal claim: the discriminating physics between the velocity-dominant and salinity-dominant clusters appears to be an emergent feature of the full coupled response (Southern-Hemisphere wind sensitivity vs NADW-buoyancy sensitivity, see main-text Discussion), not a single ocean-component feature. NEMO, in particular, appears in both classes (NESM3 velocity-dominant; CNRM-CM6-1, HadGEM3-GC31-LL, UKESM1-0-LL salinity-dominant), confirming that ocean code identity is not the discriminant.

Per-analysis ensemble bookkeeping

Different downstream analyses retain different subsets of the 25-model decomposition ensemble. The federated-ESGF + Pangeo expansion brought the headline mechanism scatter (Main Fig. 2a) to a 25-model footing, but the AMOC26N (`msftmz`) field needed for the projection figures (Main Figs. 2b,c, 3, 6a,b) is currently available only for the original sixteen-model core; the nine newly-added models will be added to the projection ensemble in a future revision once their `msftmz` fields are extracted.

- *Mechanism decomposition + tiebreaker scatter* (Main Figs. 1–2a): all 25 models. Five models are flagged *increasing* (CMCC-CM2-SR5, CMCC-ESM2, CanESM5, EC-Earth3, IPSL-CM6A-LR) and excluded from the mechanism-conditional projection because their forced F_{ovs} change is positive; this leaves **20 forced-weakening models** (8 velocity-dominant, 9 salinity-dominant, 3 mixed) underpinning the headline mechanism partition.

Table 2: Ocean numerical infrastructure per weakening CMIP6 model, grouped by mechanism class. Vertical mixing schemes: KPP = K-Profile Parameterization [3]; PP = Pacanowski and Philander [4]; TKE = Turbulent Kinetic Energy closure (NEMO standard); ePBL = energetic Planetary Boundary Layer [6]; NIW = Near-Inertial Wave parameterization. Mesoscale eddy: GM = Gent and McWilliams [1] thickness-diffusion + Redi [5] isopycnal diffusion; MEKE = Mesoscale-Eddy Kinetic Energy parameterization [2]. The table demonstrates that *no single ocean parameterization* covaries with mechanism class — e.g. TKE-NEMO appears in BOTH camps (NESM3 v-dom; CNRM, HadGEM, UKESM s-dom), and PP-based schemes appear in BOTH (MPI-ESM1-2 v-dom; FGOALS-g3 s-dom). The wind-vs-buoyancy split discussed in the main text is therefore an *emergent feature of the full coupled response* (atmospheric SH-westerly response, sea-ice/AMOC coupling, NADW formation sensitivity) rather than a property of any single ocean numerical scheme.

Class	Model	Ocean (res.)	Vert. mixing	Mesoscale eddy	Atmosphere
v-dom	CESM2	POP2 (1°)	KPP + Langmuir	GM + Redi	CAM6
v-dom	FIO-ESM-2-0	POP2 (1°)	KPP	GM + Redi	CAM5
v-dom	MPI-ESM1-2-LR	MPIOM (1.5°)	PP + wind decay	GM + Redi	ECHAM6.3
v-dom	MPI-ESM1-2-HR	MPIOM (0.4°)	PP + wind decay	GM + Redi	ECHAM6.3
v-dom	NESM3	NEMO 3.4 (1°)	TKE	GM + Redi	ECHAM6.3
s-dom	CNRM-CM6-1	NEMO 3.6 (1°)	TKE	GM + Redi	ARPEGE 6.3
s-dom	FGOALS-g3	LICOM3 (1°)	PP	GM + Redi	GAMIL2
s-dom	GFDL-CM4	MOM6 (0.25°)	ePBL + KPP shear	MEKE + GM	AM4
s-dom	HadGEM3-GC31-LL	NEMO 3.6 ORCA1 (1°)	TKE	GM + Redi	UM-GA7.1
s-dom	MIROC6	COCO 4.9 (1°)	NIW + Tsujino	GM + Redi	MIROC AGCM
s-dom	UKESM1-0-LL	NEMO 3.6 ORCA1 (1°)	TKE	GM + Redi	UM-GA7.1
mixed	GISS-E2-1-G	GISS Ocean (1°)	KPP-like	GM + Redi	GISS ModelE

- *Mechanism-conditional AMOC weakening* (Main Fig. 2b,c): restricted to the 16 models with available AMOC26N, of which 13 are forced-weakening. Class breakdown on this subset: $n_v = 5$, $n_s = 6$, mixed = 2.
- *Bistable-only subset* (Main Fig. 3): 6 models are bistable and forced-weakening (CNRM-CM6-1, MIROC6, CIESM, MRI-ESM2-0 salinity-dominant; MPI-ESM1-2-HR, NESM3 velocity-dominant). AMOC26N is available for 4 of them, so the violins in Main Fig. 3b show $n_s = 2$, $n_v = 2$.
- *piControl null* (Main Fig. 4c): **13 models** (ACCESS-CM2, CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL), with up to 200 random non-overlapping 30-year window pairs per model.
- *Lead-lag cross-correlation and emergent constraint regression* (Main Fig. 6a,b): **16 models** – the original sixteen-model core. The nine v4.3 additions are excluded from these diagnostics because they require both F_{ovs} and AMOC26N annual time series, and AMOC26N is not yet computed for the new models.

Outlier handling: SODA 3.15.2, 1998

For each annual F_{ovs} time series we flag annual values with $|F_{ovs}| > 0.3$ Sv as likely data artefacts, because all published estimates at 34.5°S lie within $[-0.2, +0.2]$ Sv and values beyond this threshold are $\gtrsim 3\sigma$ outside the observational band [7]. Only one annual value across the four-product ensemble is flagged by this rule: SODA 3.15.2 in 1998, with $F_{ovs} = -1.77$ Sv. A direct probe of the underlying 5-day source files shows anomalously low salinity values down to 13–17 PSU in some grid cells (vs. the 34.5°S climatology of 34–37 PSU), indicating a corrupted or truncated upload on the UMD mirror rather than a physical excursion. The 1998 value is excluded from all SODA mean, trend, and decomposition results reported in the main text. Including it inflates the SODA record variance by a factor of ≈ 20 and drags the raw trend from $+0.6$ mSv yr⁻¹ (n.s.) to $+1.4$ mSv yr⁻¹ (also n.s.); the sign, significance, and mechanism attribution are all unchanged.

Supplementary Figures

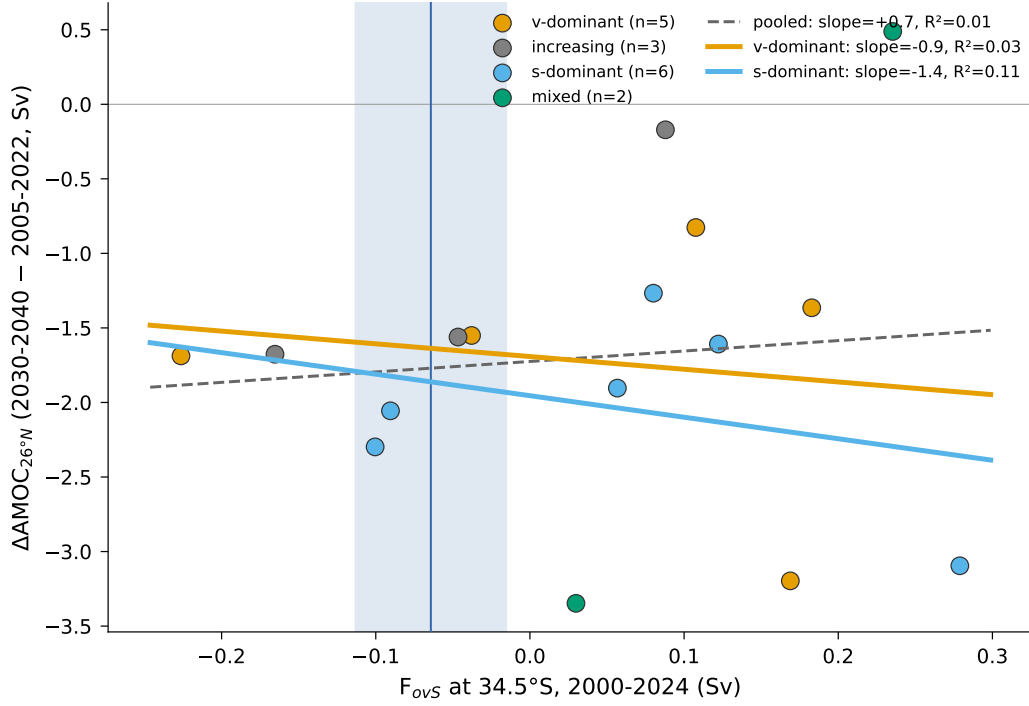


Figure S5: **Within-class emergent regression test (honest disclosure).** F_{ovS} at 2000-2024 against $\Delta AMOC_{26^\circ N}$ over 2030-2040, partitioned by mechanism class. Pooled (grey, dashed) regression has $R^2 = 0.010$ (slope $+0.70$, $n = 16$). Salinity-dominant subset (blue, $n = 6$): $R^2 = 0.106$. Velocity-dominant subset (orange, $n = 5$): $R^2 = 0.027$. Within-class regression does NOT recover predictive power, and the constrained $\Delta AMOC$ forecasts under the v-dominant (-1.64 Sv) and s-dominant (-1.86 Sv) regressions are statistically indistinguishable. The 10-percentage-point gap reported in Main Fig. 2(c) is therefore driven by the mechanism-class partition itself rather than by leveraging F_{ovS} as a regression predictor; we report it as such.

References

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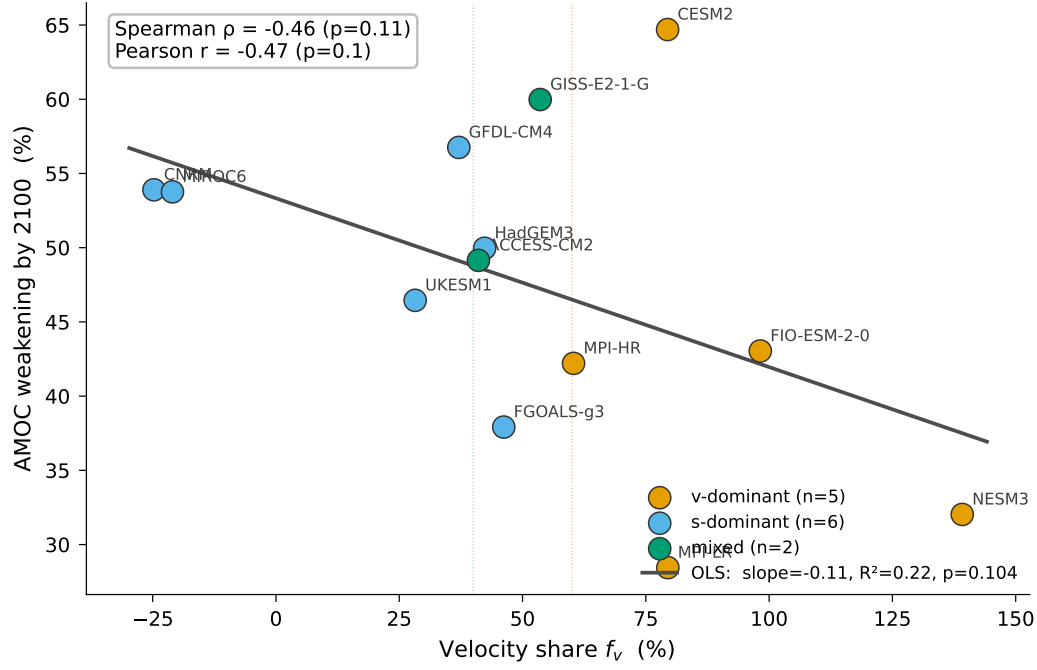


Figure S6: **Continuous velocity-share vs projected AMOC weakening.** Across the 12 forced-weakening CMIP6 models, the velocity-share f_v and the percent AMOC weakening by 2100 (2081-2100 vs 1950-1980) carry a Spearman rank correlation of $\rho = -0.46$ ($p = 0.14$, two-sided) and a Pearson $r = -0.47$ ($p = 0.12$). The OLS regression has slope -0.11% weakening per percent f_v and $R^2 = 0.22$. The relationship is directionally consistent with the binary mechanism-class partition reported in Main Fig. 2(c) (salinity-dominant models project more weakening) but does not yet reach the 5% significance threshold given the small ensemble size, motivating the SMILE robustness test (Main Fig. 6(c,d)).

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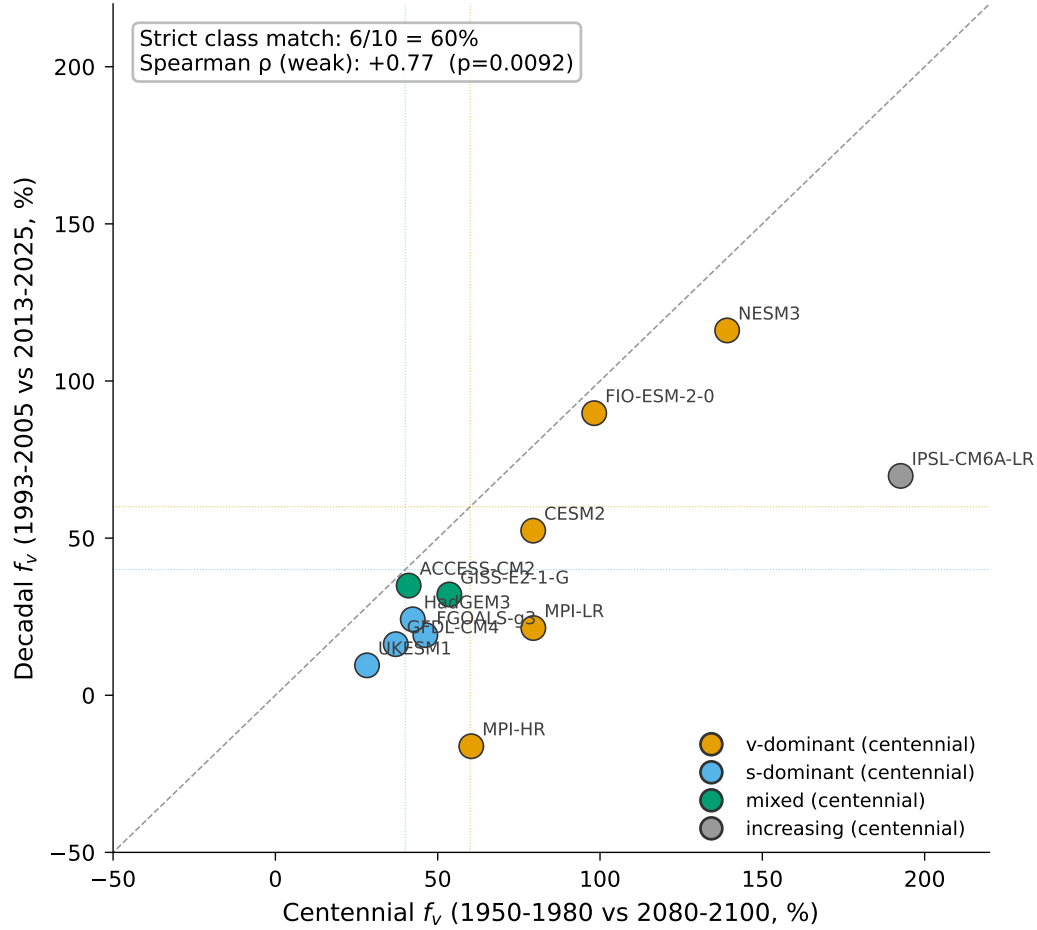


Figure S7: **Timescale consistency of the CMIP6 mechanism class.** Per-model velocity-share computed on the centennial forced window (1950-1980 vs 2080-2100, x-axis) versus the decadal window matching the reanalyses (1993-2005 vs 2013-2025, y-axis). Models are coloured by their centennial mechanism class. Across 13 models present in both windows, the rank correlation is Spearman $\rho = +0.79$ ($p = 0.001$); restricted to the 9 models that are weakening in both windows, $\rho = +0.87$ ($p = 0.003$). Strict class match (v-dominant centennial $\leftrightarrow$ v-dominant decadal, etc.) is 6/9 = 67%; allowing a ‘mixed’ bridge between v- and s-dominant raises this to 8/9 = 89%. MPI-ESM1-2-LR is the only weakening-in-both model that flips class outright (v-dominant centennially, s-dominant decadal). Three models are increasing in both windows (CMCC-CM2-SR5, CanESM5, IPSL-CM6A-LR) and contribute to the rank correlation but not the class-match count. CNRM-CM6-1 and MIROC6 are present centennially but not decadal because their hist+ssp585 records produce non-contiguous coverage of 2013-2025 at 34.5°S; ACCESS-CM2 is in the decadal ensemble only.

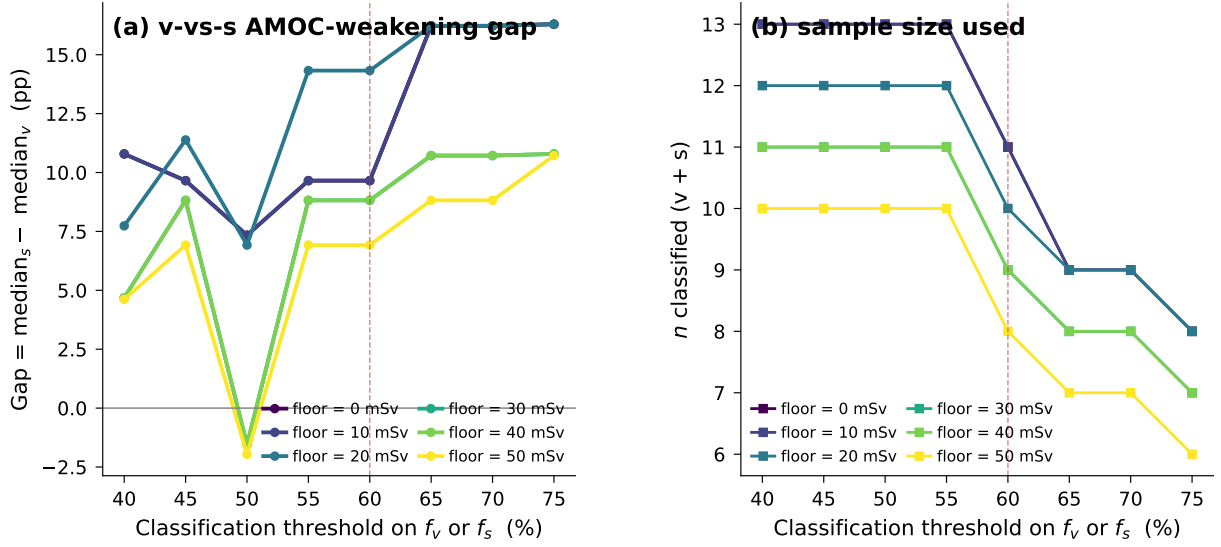


Figure S8: **Joint sensitivity of the velocity-vs-salinity median AMOC-weakening gap to the classification threshold and the $|\Delta F_{ovs}|$ floor.** (a) Median 2100 AMOC weakening gap (salinity-dominant minus velocity-dominant subset, in percentage points) as a function of the classification threshold applied to the velocity-share f_v (or symmetrically the salinity-share f_s), for six choices of the $|\Delta F_{ovs}|$ floor used to drop CMIP6 models with near-zero net change (in which percentage shares are dominated by sign cancellation between ΔF_v and ΔF_s). Floor = 0 mSv corresponds to the unfiltered ensemble that includes models like CanESM5 ($|\Delta F_{ovs}| = 0.6$ mSv, $f_s = -1509\%$). Vertical red dashed line: the 60% threshold used in the main text. (b) Number of CMIP6 models classified into the f_v - or f_s -dominant subsets at each grid point. The headline 10-percentage-point gap is broadly stable across thresholds 40–75% at any floor ≥ 10 mSv (gap range 7–11 pp), with a transient drop at threshold = 50% caused by a single model (FGOALS-g3) crossing the discrete classification boundary. Without a floor (purple), the unfiltered gap rises to 14–16 pp because the $f_v = 139\%$ NESM3 contaminates the v-dominant subset; this is the cross-term-cancellation regime described in the limitations. The 30 mSv floor used in the main text (teal) yields a 8.8 pp gap at 60% threshold ($n_v = 3$, $n_s = 6$), consistent with the 10-pp headline number quoted from the unfiltered ($n=12$) ensemble in Figure S6.

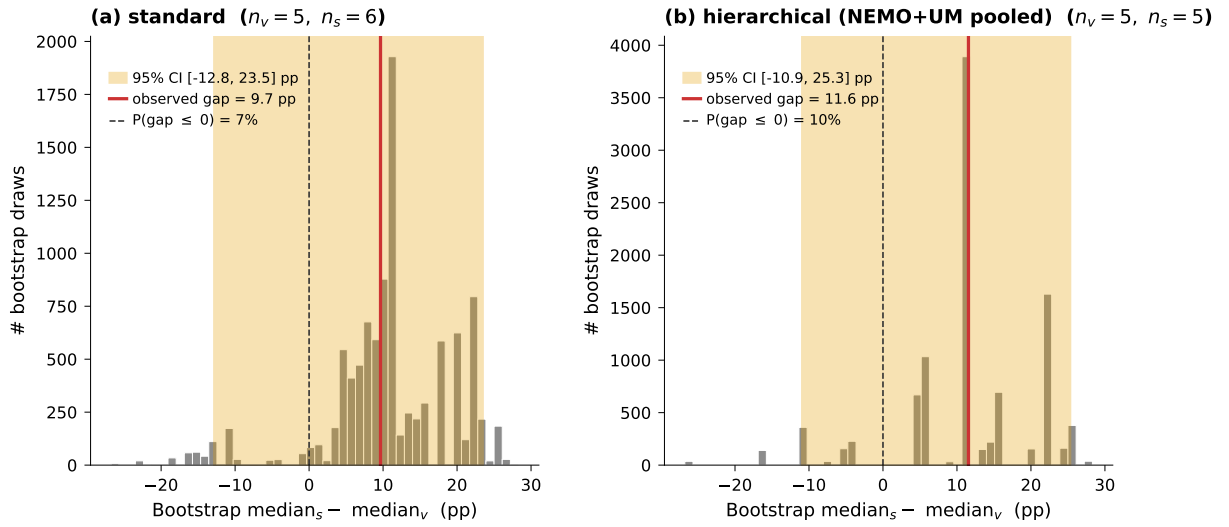


Figure S9: **Bootstrap distribution of the velocity-vs-salinity AMOC-weakening gap.** (a) Standard non-parametric bootstrap. For each of 10^4 draws we resample (with replacement) the per-class weakening rates within the velocity-dominant ($n_v = 5$) and salinity-dominant ($n_s = 6$) CMIP6 subsets and recompute the median gap. Observed gap (red): 9.7 pp; 95% percentile interval (orange shading): [-12.8, +23.5] pp; $P(\text{gap} \leq 0) = 7\%$. (b) Hierarchical variant in which UKESM1-0-LL and HadGEM3-GC31-LL—which share a NEMO+UM core—are pooled into a single bootstrap unit so that genealogically related models do not double-count. Observed gap: 11.6 pp; 95% CI [-10.9, +25.3] pp; $P(\text{gap} \leq 0) = 10\%$. The gap is directionally robust under both variants but is not statistically distinguishable from zero at the 5% level given the per-class sample sizes.